**1** **(a)** $80 \times 45 = 3600$ **[M1]**

0.36 **[A1]**

**(b)** 3200 **[B1]**

*[Total Marks: 3]*

**2** **(a)** 35000 **[B1]**

**(b)** $60 \times 45 \times 30$ OR $2700 \times 30$ OR 81000 **[M1]**

their $81000 \div 1000$ **[M1]**

81 **[A1]**

*[Total Marks: 4]*

**3** **(a)** Converts dimensions to consistent units, e.g. 150 cm, 80 cm, 60 cm OR 1.5 m, 0.8 m, 0.6 m **[M1]**

$150 \times 80 \times 60 = 720000$ OR $1.5 \times 0.8 \times 0.6 = 0.72$ **[M1]**

720 **[A1]**

**(b)** $0.024 \times 1000000 = 24000$ **[M1]**

Mass $= 24000 \times 7.8 = 187200$ **[M1]**

187.2 **[A1]**

*[Total Marks: 6]*

**4** **(a)** Conversion of units: $15\ mm = 0.015\ m$ OR $15\ m = 1500\ cm$ etc. **[M1]**

Volume $= 15 \times 12 \times 0.015$ (or eq) **[M1]**

2700 **[A1]**

**(b)** Use of consistent units for tank volume equation (e.g. $r = 0.6\ m$ and $V = 2.7\ m^3$ OR $V = 2700000\ cm^3$ and $r = 60\ cm$) **[M1]**

$\pi \times 0.6^2 \times h = 2.7$ OR $\pi \times 60^2 \times h = 2700000$ **[M1]**

$h = \dfrac{2.7}{0.36\pi}$ (m) OR $h = \dfrac{2700000}{3600\pi}$ (cm) **[M1]**

238.7 **[A1]**

*[Total Marks: 7]*

**5** **(a)** Identifies consistent units for radius and length (e.g. $r = 1.5$ cm and length = 80 cm) **[M1]**

Volume in one second $= \pi \times 1.5^2 \times 80 = 180\pi\ cm^3$ $(= 565.48\ldots\ cm^3)$ **[M1]**

Converts to litres $(180\pi \div 1000)$ and confirms $0.56548\cdots \approx 0.565$ **[A1]**

**(b)** Volume of pool required $= 4 \times 2.5 \times 1.2 = 12\ m^3$ OR 12000 litres **[M1]**

Time in seconds $= 12000 \div 0.56548\ldots$ OR $12000 \div \dfrac{180\pi}{1000}$ **[M1]**

$= 21220.6\ldots$ seconds ($353.6\ldots$ minutes) **[A1]**

5 hours 54 minutes **[A1]**

*[Total Marks: 7]*

**6 (a)** Cross-sectional area $= \pi \times 2.5^2$ $(= 6.25\pi)$ **[M1]**

Speed conversion: $18\ km/h = \dfrac{18 \times 1000 \times 100}{60}\ cm/min$ $(= 30000\ cm/min)$ **[M1]**

Volume per minute $= 6.25\pi \times 30000$ $(= 187500\pi\ cm^3)$ **[M1]**

Convert to litres: $\dfrac{187500\pi}{1000}$ **[M1]**

589 or 589.0 to 589.1 $(187.5\pi)$ **[A1]**

**(b)** Capacity of reservoir $= 50 \times 30 \times 4 = 6000\ m^3$ OR 6000000 litres **[M1]**

Time in minutes $= \dfrac{6000000}{187.5\pi}$ (or 6000000÷ their (a)) $(= 10185.9...)$ **[M1]**

Time in hours $= \dfrac{10185.9...}{60}$ $(= 169.76...)$ **[M1]**

$169.76... \div 24 = 7.07...$ days so 7 full days **[M1]**

$0.07... \times 24 = 1.76...$ hours (or $169.76... - 7 \times 24$) **[M1]**

7 **[A1]**

2 **[A1]**

*[Total Marks: 12]*

**7 (a)** 5.2 **[B1]**

**(b)** 4700 **[B1]**

**(c)** 850 **[B1]**

*[Total Marks: 3]*

**8 (a)** Converts all dimensions to cm: 250, 140, 0.8 **[M1]**

$250 \times 140 \times 0.8 = 28000$ **[A1]**

**(b)** Multiplies volume by 7.8: $28000 \times 7.8 = 218400$ **[M1]**

218.4 **[A1]**

*[Total Marks: 4]*

**9 (a)** Converts speed to cm/min: $5.4 \times 1000 \times 100 \div 60 = 9000$ **[M1]**

Calculates volume in 1 minute: $\pi \times 2.5^2 \times 9000 = 56250\pi\ cm^3$ **[M1]**

Converts $cm^3$ to litres and evaluates: $56250\pi \div 1000 = 176.714\ldots$ **[A1]**

**(b)** Calculates total volume in 45 minutes: $56250\pi \times 45 = 2531250\pi\ cm^3$ (or uses litres: $176.714\cdots \times 45 = 7952.1\ldots$ litres) **[M1]**

Converts tank base dimensions to cm and finds area: $300 \times 200 = 60000\ cm^2$ **[M1]**

Sets up equation for depth: Depth $= \dfrac{2531250\pi}{60000}$ **[M1]**

133 (or 132.5 to 132.6) **[A1]**

*[Total Marks: 7]*